



Voice AI Analytics

Audio Intelligence Pipeline & Conversational Data Interface

for Heya AI · Technical R&D Proposal

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1. Executive Summary

1.1 The Problem

Heya AI deploys AI-powered voice agents that handle inbound and outbound calls for Australian small-to-medium businesses. Each call generates an audio recording and a full transcript, yet the current client dashboard surfaces only coarse metadata — call volume, Retell-sourced sentiment (positive / neutral / negative), disconnection reason, and success rate.

Audio recordings remain entirely unanalysed. As a result, critical behavioural and emotional signals — tone, pace, hesitation, interruptions, and vocal energy — are invisible to clients. Non-technical business owners have no way to query their call data in natural language, forcing reliance on manual review or developer support.

Core pain point: the absence of audio-derived insights that would enable richer, actionable analytics and empower non-technical users to interrogate their call data conversationally.

1.2 The Solution

This project delivers two tightly integrated components:

- **Audio Intelligence Pipeline:** a Speech AI layer (SpeechBrain ^[3] + emotion2vec ^[2]) that processes raw call recordings to extract acoustic emotional and behavioural signals — MFCCs, pitch, energy, speaking rate, pause patterns, and composite engagement metrics — storing results alongside existing metadata in the Heya AI database.
- **Conversational Data Interface:** a Retrieval-Augmented Generation (RAG) system ^[1] that allows non-technical users to query transcripts, metadata, and audio-derived insights in natural language via a secure REST API with enforced per-client tenant scoping.

Together, these components close the gap between raw call data and actionable business intelligence — providing capabilities unavailable in any existing competitor platform.

2. Technical Objectives

Objective	Description
Acoustic Intelligence	Quantitative measurement of per-call engagement, emotion trajectory, speaker dynamics, hesitation frequency, and interruption events — derived solely from acoustic signals, not transcripts.
Actionable Insights	A natural-language conversational interface enabling non-technical business owners to query their call data without SQL or developer support, with grounded, hallucination-free responses.

Objective	Description
Security & Privacy	Strict per-client tenant scoping enforced at the API and database layers; full compliance with the Australian Privacy Principles (Privacy Act 1988); transient processing with immediate deletion of raw audio post-analysis.

3. Proposed System Architecture

The diagram below illustrates the end-to-end system, combining the RAG-based Conversational Data Interface (left) with a Key Concepts Mind Map (right) that maps each architectural component to its business purpose.

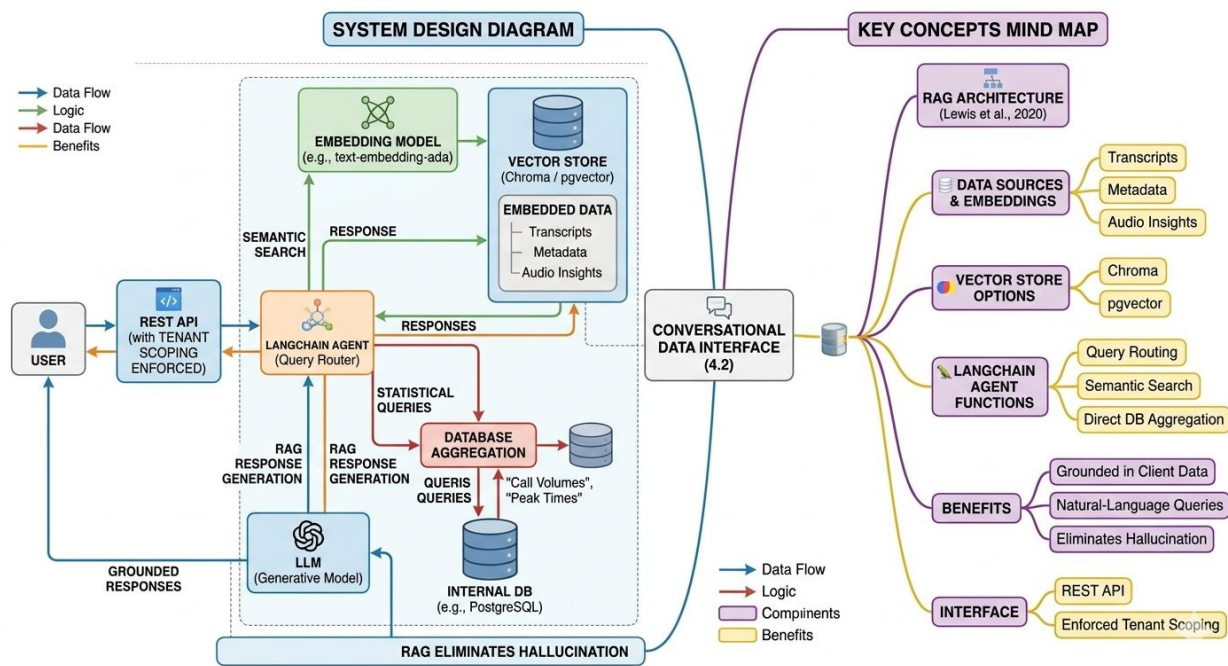


Figure 1: System Design Diagram and Key Concepts Mind Map — Heya AI Voice Analytics Platform

3.1 Phase A — Audio Processing Pipeline

3.1.1 Pre-Processing & Feature Extraction

Raw call recordings (variable-length audio, typically 1–10 minutes, Australian accent) are ingested and pre-processed using the SpeechBrain toolkit [9] (Ravanelli et al., 2021) in combination with librosa. The following acoustic feature classes are extracted per speaker turn:

- MFCCs (Mel-Frequency Cepstral Coefficients) — spectral fingerprint of speech articulation. [2, 13]
- Pitch (F0) — fundamental frequency contour, indicative of emotional arousal. [2, 3]
- Energy (RMS) — amplitude envelope, correlating with emphasis and stress. [2, 12]
- Speaking rate — syllables per second, capturing pace and urgency. [3, 11]
- Pause patterns — duration and frequency of silence, reflecting hesitation. [3, 11]

3.1.2 Emotion Classification

Per-speaker emotion classification at turn and segment level is performed using emotion2vec ^[7] (Ma et al., 2024), a self-supervised universal speech emotion representation model. emotion2vec is pre-trained on diverse speech corpora and fine-tuned for deployment on variable-quality, Australian-accented recordings — delivering high classification accuracy without requiring large labelled datasets.

3.1.3 Composite Metrics

Rule-based acoustic analysis combined with lightweight models computes three composite business metrics per call:

- **Engagement Score** — a weighted aggregate of energy, speaking rate, and response latency.
- **Hesitation Markers** — frequency and duration of filled pauses ('um', 'uh') and silence gaps.
- **Interruption Events** — instances of overlapping speech detected via diarization.

All extracted insights are stored alongside existing JSON metadata in the Heya AI database, enabling downstream querying and dashboard visualisation.

3.2 Phase B — Data Storage & Retrieval

Transcripts, existing metadata, and the newly generated audio insights are embedded into a vector store using a text embedding model (e.g., text-embedding-ada-002 or equivalent open-source alternative).

Two vector store options are evaluated:

- **Chroma** — lightweight, local-first, suitable for development and smaller deployments. ^[5]
- **pgvector** — PostgreSQL extension, preferred for production due to its integration with the existing Heya AI relational database and support for row-level security. ^[6]

The embedded data corpus comprises transcripts, call metadata (call ID, timestamp, duration, existing sentiment), and the three new audio-derived metrics per call.

3.3 Phase C — Conversational Interface

A LangChain agent (Query Router) mediates between user intent and the appropriate data retrieval strategy:

- **Semantic Search path:** for topic, theme, or sentiment queries — the agent embeds the user's natural-language question and retrieves the most relevant chunks from the vector store, then passes them to the LLM (Generative Model) for a grounded response.
- **Direct DB Aggregation path:** for statistical queries such as 'How many calls did I receive last Tuesday?' or 'What were peak call times this month?' — the agent generates and executes a parameterised SQL query directly against the internal database, bypassing the LLM for structured outputs.

The interface is exposed as a REST API with enforced tenant scoping. Every query is bound to the authenticated client's data partition, preventing any cross-tenant data leakage. Responses are returned with references to the source data, enabling auditability.

4. Methodology & Justification

4.1 Why emotion2vec?

emotion2vec is a self-supervised model that learns speech emotion representations from unlabelled audio — critical for Australian-accented, variable-quality recordings where large labelled datasets do not exist.

Traditional supervised emotion classifiers require large, high-quality, labelled datasets in the target accent. ^[4] Heya AI's call recordings are Australian-accented, of variable quality (telephony compression, background noise), and arrive without emotion labels. emotion2vec addresses this by:

- Self-supervised pre-training on diverse speech corpora, providing robust generalisation across accents and audio conditions. ^[7]
- Universal representation learning — a single model handles multiple emotion categories without per-category retraining. ^[7]
- State-of-the-art performance on benchmark emotion recognition datasets (IEMOCAP, MSP-Podcast) with minimal fine-tuning data (Ma et al., 2024). ^[7]

A fallback mechanism is built in: if emotion2vec classification confidence falls below a defined threshold, the system reverts to rule-based acoustic heuristics (energy + pitch patterns), ensuring the minimum two-insight requirement is always met.

4.2 Why RAG?

RAG grounds every LLM response in the client's own data — eliminating hallucinations while enabling natural-language queries over structured business intelligence.

Standard LLMs generate responses from parametric memory, making them unsuitable for queries against proprietary, time-sensitive call data. RAG solves this by:

- **Retrieval grounding:** every response is generated from retrieved excerpts of the client's actual transcripts, metadata, and audio insights — not from model weights. ^[6]
- **Hallucination elimination:** if no relevant data is found, the system returns a no-data response rather than fabricating an answer. ^[6, 10]
- **Dynamic knowledge base:** as new calls are ingested and embedded, the knowledge base updates automatically — no model retraining required. ^[6]
- **Multi-tenant isolation:** the vector store and DB aggregation paths both enforce client-level partitioning, ensuring data sovereignty. ^[5]

The hybrid routing strategy (semantic search + direct DB aggregation) ensures optimal performance across both qualitative and quantitative query types, matching the full range of questions a call centre manager might ask.

4.3 Frameworks & Reproducibility

SpeechBrain ^[9] (Ravanelli et al., 2021) provides a unified, reproducible framework for speaker diarization and acoustic feature extraction. Its modular architecture enables straightforward substitution of components as superior models emerge. The full Python stack — SpeechBrain, Hugging Face Transformers, librosa, LangChain, FastAPI — is open-source and runs on standard cloud instances, with no proprietary hardware dependencies.

5. Implementation Roadmap

The project is divided into four delivery stages, executed across a 9-week build sprint using an Agile framework with 2-week iterations and weekly internal stand-ups.

Stage 1 — Data Ingestion & Anonymisation (Weeks 1–2)

1. Receive and profile the anonymised dataset supplied by Heya AI.
2. Validate audio file integrity, metadata schema, and transcript alignment.
3. Establish data pipeline: ingestion → anonymisation check → transient storage → feature extraction queue.
4. Finalise and obtain sign-off on the Technical Design Document (architecture, model selection, schema changes).

Stage 2 — Feature Engineering & Emotion Model (Weeks 2–4)

5. Implement SpeechBrain diarization pipeline; validate speaker segmentation on sample calls.
6. Extract MFCCs, pitch, energy, speaking rate, and pause patterns using librosa.
7. Deploy and evaluate emotion2vec on hold-out call sample; tune confidence threshold for fallback.
8. Compute composite metrics (engagement score, hesitation markers, interruption events).
9. Integrate outputs into the Heya AI database schema; deliver Pipeline v1 tested on 200 calls.

Stage 3 — Vector Database & RAG Development (Weeks 5–6)

10. Embed transcripts, metadata, and audio insights into the selected vector store (Chroma → pgvector for production).

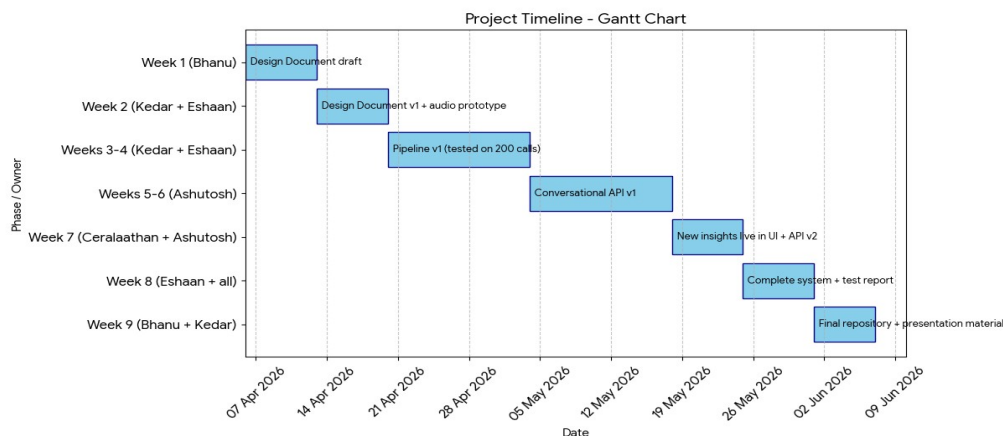
11. Implement LangChain agent with semantic search and DB aggregation routing.
12. Build and expose REST API with tenant-scoped authentication; enforce row-level data isolation.
13. Validate query accuracy against the automated test suite (SOW example queries).

Stage 4 — API Exposure & Frontend Integration (Weeks 7–8)

14. Integrate new audio metrics into the Heya AI client dashboard with at least one new visualisation.
15. Optimise API response latency to ≤ 10 seconds under representative query load.
16. Conduct comprehensive end-to-end testing, privacy audit, and cross-tenant penetration testing.
17. Deliver complete system and test report; finalise documentation and handover materials.

5.1 Detailed 9-Week Timeline

Week	Dates	Stage	Key Tasks	Owner
1	6–12 Apr	Stage 1	Dataset exploration, literature review, initial architecture design	Bhanu
2	13–19 Apr	Stage 2	Model selection, prototype audio features & SER	Kedar + Eshaan
3–4	20 Apr–3 May	Stage 2	Full audio pipeline, DB integration, diarization testing	Kedar + Eshaan
5–6	4–17 May	Stage 3	Vector store population, RAG agent, query routing	Ashutosh
7	18–24 May	Stage 4	Dashboard integration, metric visualisations, end-to-end testing	Gnanaprakash + Ashutosh
8	25–31 May	Stage 4	Comprehensive testing, optimisation, security & privacy audit	Eshaan + all
9	1–6 Jun	Handover	Final documentation, README, handover slides, rehearsal	Bhanu + Kedar



6. Ethical Considerations & Data Governance

6.1 Anonymised Dataset Handling

All call recordings and associated metadata supplied by Heya AI are fully anonymised prior to project ingestion. Personally identifiable information (caller names, phone numbers, account references) is removed at source. Within the project pipeline:

- Raw audio files are processed transiently — stored only for the duration of feature extraction, then deleted immediately.
- Derived features (MFCCs, emotion scores, composite metrics) contain no personally identifiable information and are safe for persistent storage.
- All team members operate under a data handling agreement aligned with Heya AI's existing privacy framework.

6.2 Tenant Scoping in Multi-Tenant Environments

Heya AI serves multiple SMB clients on a shared infrastructure. The conversational interface enforces strict data isolation through two complementary mechanisms:

- **API-layer scoping:** every query request carries a verified client token; the LangChain agent's vector store retrieval and DB aggregation queries are parameterised with the authenticated client's partition identifier.
- **Database-layer row-level security:** PostgreSQL row-level security policies ensure that even if an application-layer bug exists, a client cannot access another tenant's data at the storage level.

The automated test suite includes a dedicated cross-tenant isolation test battery that must pass with zero leakage before any deployment.

6.3 Regulatory Compliance

The project operates under the following regulatory frameworks:

- **Australian Privacy Principles (Privacy Act 1988):** the primary compliance framework. All data processing is transient; no raw audio is retained post-analysis; data minimisation principles are applied throughout. ^[1, 8]
- **GDPR alignment (where applicable):** Heya AI may serve businesses with European customers; the pipeline design respects data minimisation, purpose limitation, and the right to erasure.
- **Institutional ethics:** the project uses only data supplied directly by Heya AI under an existing commercial relationship; no public data collection or human participant research is involved.

7. Expected Outcomes & Return on Investment

The following outcomes are expected upon successful delivery of the minimum viable aims, with measurable indicators aligned to the Statement of Work:

Stakeholder	Expected Outcome	Measurable Indicator
Call Centre Managers	Reduced time-to-insight from call data — no SQL required	Query response ≤ 10 s; NL interface accuracy 100%
Business Owners	Identification of peak call times and high-conflict interactions	≥ 2 new dashboard metrics live post-deployment
Heya AI (Product)	Competitive differentiation in the Australian SMB voice-agent market	Acoustic insights unavailable in existing competing platforms
Heya AI (Engineering)	Reproducible, extensible audio pipeline for future model upgrades	Full documentation + clean Git repository
End Customers	Improved service quality driven by richer agent feedback loops	Engagement score and hesitation markers surfaced per call

7.1 Competitive Advantage for Heya AI

No integrated solution currently combines a comprehensive audio-intelligence layer with an embedded natural-language conversational query interface tailored to a custom voice-agent platform. This project gives Heya AI a first-mover advantage in the Australian SMB market — a gap that broader platforms in the industry do not address for this segment.

7.2 Risk Management

Risk	Likelihood	Impact	Mitigation
Model accuracy shortfalls on Australian-accented audio	Medium	High	Fallback to rule-based metrics; still satisfies minimum two-insight requirement
Dataset quality / volume insufficient	Low	High	Early data profiling in Week 1; flag to Abraham before build commences
RAG latency exceeds 10-second SLA	Medium	Medium	Query caching, index optimisation; API v2 milestone in Week 7
Scope creep beyond SOW	Medium	Medium	Strict sprint gating; stretch aims deferred until core deliverables are signed off
Data privacy / tenant leakage	Low	Critical	Row-level security enforced at DB layer; automated cross-tenant test suite

8. Team, Roles & Deliverables

8.1 Team Roles

- **Testing & Evaluation Lead:** Eshaan Abhay Patil — end-to-end testing, accuracy evaluation, security audit, and stretch-aim implementation.
- **Dashboard & Visualisation Lead:** Ceralaathan Gnanaprakash — frontend integration of audio insights and conversational interface.
- **Scrum Master & Audio Pipeline:** Kedar Avinash Kubal — overall coordination, feature extraction, emotion classification implementation, and pipeline integration.
- **Conversational Interface Lead:** Ashutosh Panda — RAG architecture, vector store population, LangChain agent development, and API endpoint.
- **Documentation & Compliance Lead:** Bhanuchandu Bochu — Technical Design Document, privacy/compliance audit, final README, and handover materials.

All members contribute to integration testing and sprint reviews. Dependencies are explicit: audio pipeline outputs feed the RAG component; RAG outputs feed dashboard visualisations.

8.2 Deliverables Summary

#	Deliverable	Success Criteria
1	Technical Design Document	Architecture, model rationale, data flow and schema changes — approved before build commences
2	Audio Intelligence Pipeline	≥2 audio-derived insights per call; ≥80% emotion classification agreement on hold-out test set
3	Conversational Interface API	REST endpoint responding within 10 s; 100% accuracy on automated test suite; zero cross-tenant data leakage
4	Dashboard Frontend Components	New audio metrics live in client dashboard with at least one new visualisation
5	Git Repository & Documentation	Clean repository with full documentation and deployment README
6	Handover Presentation & Demo	Live demonstration at Showcase (7 June 2026)

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